Generating Daylight Images from Night/Low-Light Images Using Deep Learning

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***Abstract*—Low-light and night-time images often suffer from poor visibility, low contrast, and loss of details, making them unsuitable for various computer vision applications such as surveillance, autonomous driving, and image analysis. Traditional image enhancement techniques generally improve brightness but fail to generate realistic daytime appearances from night images. This project presents a deep learning-based image-to-image translation approach using the Pix2Pix Generative Adversarial Network (GAN) to convert low-light images into daylight images.**

The proposed model employs a conditional GAN architecture consisting of a U-Net based generator and a PatchGAN discriminator. The model is trained on paired night-day image datasets where corresponding daytime images act as ground truth references. The generator learns to produce realistic daylight images, while the discriminator distinguishes between real and generated images, improving the overall image quality.

Experimental results demonstrate that the Pix2Pix model effectively enhances image brightness, color consistency, and scene visibility while preserving structural details. The generated outputs achieve satisfactory visual quality and quantitative performance based on evaluation metrics such as Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM). The proposed approach can be further extended for real-time low-light image enhancement and various computer vision applications.

***Index Terms*—Deep Learning, Generative Adversarial Networks, Pix2Pix, Image-to-Image Translation, Low-Light Enhancement, Computer Vision.**

I. Introduction

Images captured during night-time or under poor lighting conditions generally contain insufficient illumination, low contrast, noise, and missing details. Such degradations negatively affect the performance of image processing and computer vision systems including object detection, facial recognition, and autonomous navigation.

Traditional enhancement techniques such as histogram equalization, gamma correction, and Retinex-based methods improve brightness but often introduce noise and fail to reconstruct realistic daylight appearances.

Recent advances in Deep Learning and Generative Adversarial Networks (GANs) have enabled significant progress in image-to-image translation tasks. Among these methods, Pix2Pix has shown remarkable success in translating images from one domain to another using paired datasets.

In this project, a Pix2Pix model is developed to transform low-light and night-time images into daylight images. The proposed system learns the mapping between night and daytime scenes and generates visually realistic outputs while preserving the structural information of the original image.

II. Related Work

Several techniques have been proposed for low-light image enhancement and image translation. Histogram Equalization (HE) improves image contrast by redistributing intensity values; however, it often causes over-enhancement and loss of details.

1) Retinex-based methods estimate illumination and reflectance components to improve visibility, but their performance degrades in complex scenes.
2) Deep learning approaches such as LLNet and EnlightenGAN have shown significant improvements in low-light enhancement tasks .
3) Furthermore, Generative Adversarial Networks (GANs) have achieved remarkable success in image generation and image-to-image translation applications. Among these methods, Pix2Pix, proposed by Isola et al.
4) Performs supervised image translation using paired datasets and has demonstrated excellent performance in generating realistic images.

III. Problem Statement and Objectives

A. Problem Statement

Images captured under low-light conditions suffer from several limitations:

1) Poor visibility and low contrast.
2) Loss of texture and structural details.
3) Presence of noise and color distortion.
4) Reduced performance of computer vision applications.

Therefore, the objective of this project is:

*Worked as Interns.

To develop a deep learning-based system using the Pix2Pix Generative Adversarial Network capable of generating realistic daylight images from night or low-light images while preserving structural information.

B. Objectives

1) To study low-light image enhancement techniques.
2) To implement Pix2Pix and CycleGAN models.
3) To generate daylight images from night and low-light images.
4) To evaluate the models using PSNR and SSIM.
5) To compare the performance of both models.

IV. Proposed Methodology

The proposed methodology consists of the following stages:

1) Dataset Collection
2) Data Preprocessing
3) Pix2Pix Model Development
4) Model Training
5) Image Generation
6) Performance Evaluation

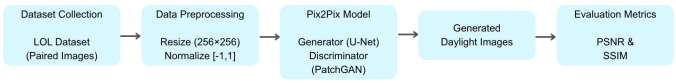


Fig. 1: Overall workflow of the proposed system.

V. Dataset Details

The proposed model is trained using the LOw-Light (LOL) dataset, a publicly available paired image dataset designed for low-light image enhancement tasks. The dataset contains 500 paired images consisting of low-light input images and their corresponding normal-light ground truth images. Out of these, 485 image pairs are used for training and 15 image pairs are used for testing. The paired nature of the dataset makes it suitable for training the Pix2Pix conditional GAN model.

TABLE I: Statistics of the dataset used in this work.

Dataset	Number of Images	Dataset Type	Purpose
LOL Dataset	500 paired images (485 training, 15 testing)	Paired low-light and normal-light images	Pix2Pix training and evaluation

Preview of some LOL Dataset Samples are shown in in Fig. 2.

VI. Data Preprocessing

The preprocessing stage plays a crucial role in improving the performance and stability of the proposed model. All images are resized to 256×256 pixels and normalized to the range [-1,1]. The paired low-light and normal-light images are then converted into tensors before being fed into the network. The overall preprocessing procedure adopted in this work is illustrated in Fig. 3.

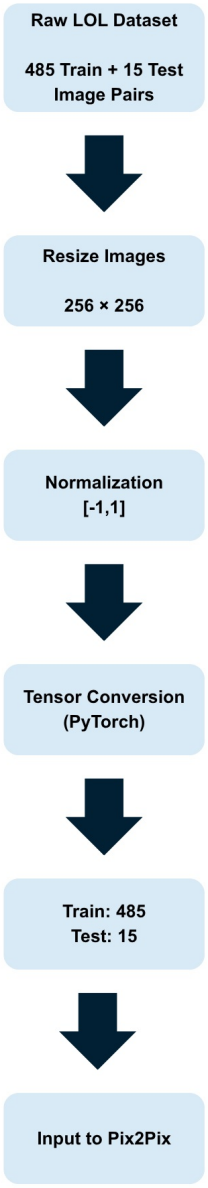


Fig. 3: Data preprocessing pipeline including resizing, normalization, and augmentation steps.



Fig. 2: Preview of some LOL Dataset Samples.

VII. Model Architecture

The proposed system is based on two Generative Adversarial Network (GAN) architectures: Pix2Pix for paired image-to-image translation and CycleGAN for unpaired image translation. Both models consist of a generator and a discriminator network trained in an adversarial manner to produce realistic daylight-like images from low-light inputs.

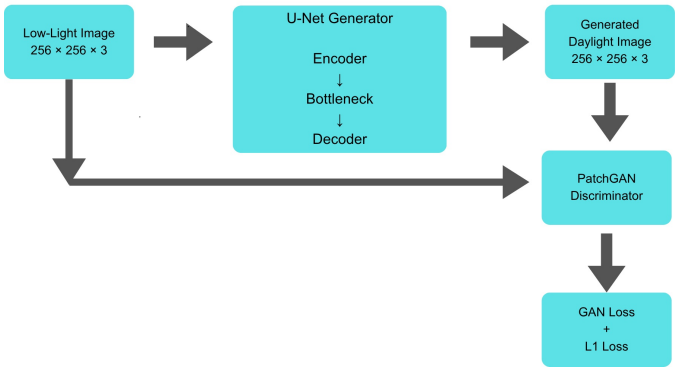


Fig. 4: Architecture of the proposed Pix2Pix model.

A. Pix2Pix Architecture

Pix2Pix is a conditional Generative Adversarial Network (cGAN) designed for paired image-to-image translation tasks. In this work, the generator learns a mapping from low-light images to corresponding normal-light images. The generator typically follows a U-Net architecture, which helps preserve fine-grained spatial details through skip connections.

The discriminator is a PatchGAN, which classifies whether local image patches are real or generated, ensuring high-frequency detail preservation in the output images. The model is trained using a combination of adversarial loss and L1 reconstruction loss, which encourages both realism and structural similarity with the ground truth images.

$$L_{Pix2Pix} = L_{cGAN} + \lambda L_{L1} \tag{1}$$

VIII. Experimental Setup

The experimental setup defines the hardware, software, and training configurations used for implementing and evaluating the proposed Pix2Pix model.

The implementation is carried out using the PyTorch deep learning framework on Google Colaboratory. The models are trained using an NVIDIA Tesla T4 GPU, which provides sufficient computational power for training GAN-based architectures.

The Adam optimizer is employed for both the generator and discriminator networks due to its efficient convergence and stability during adversarial training. A learning rate of 0.0002 with $\beta_1 = 0.5$ is used throughout the training process. The batch size is set to 16, and all input images are resized

to 256×256 pixels before being fed into the network. The proposed Pix2Pix model is successfully trained for 20 epochs on the LOL dataset and demonstrates stable convergence during training. The generated results exhibit significant improvements in brightness, color consistency, and structural details, producing realistic daylight images from low-light inputs. The performance of the model is evaluated using both qualitative analysis and quantitative metrics, namely Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM).

TABLE II: Experimental Setup

Parameter	Value
Platform	Google Colab
GPU	NVIDIA Tesla T4
Programming Language	Python 3
Framework	PyTorch
Optimizer	Adam
Learning Rate	0.0002
Batch Size	16
Epochs	20
β_1	0.5
Image Size	256×256

IX. Training Results

The training performance of the proposed Pix2Pix model is analyzed using the generator and discriminator loss curves. These loss values provide insights into the convergence behavior and stability of the adversarial training process.

During training, the generator loss gradually decreases over the epochs, indicating that the generator progressively learns to produce realistic daylight images from low-light inputs. Similarly, the discriminator loss initially fluctuates but eventually stabilizes as the generator improves its ability to generate high-quality images. The stabilization of both losses demonstrates a balanced adversarial learning process between the generator and discriminator networks.

The loss curves provide a visual representation of the model's convergence and confirm that the Pix2Pix architecture successfully learns the mapping between low-light and normal-light image domains. After 20 epochs of training, the model produces visually realistic daylight images with improved brightness, color consistency, and preservation of structural details.

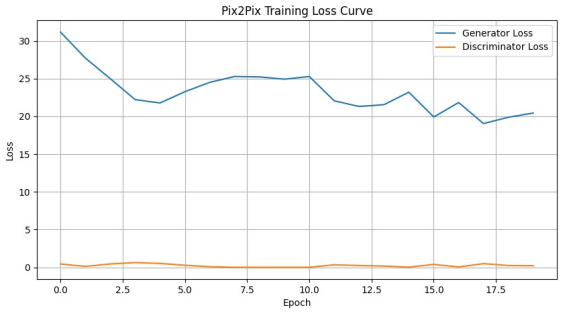


Fig. 5: Training loss curve of the Pix2Pix model.

X. Generated Results

This section presents the qualitative results of the proposed Pix2Pix model for generating daylight images from unseen low-light inputs. The generated outputs are visually analyzed in terms of brightness enhancement, color consistency, and preservation of structural details.

The Pix2Pix model produces realistic daylight images with improved illumination and better visibility while preserving important scene information. Although minor artifacts may appear in a few challenging cases, the overall results demonstrate the effectiveness of the proposed approach for low-light image enhancement and daylight image generation.

Qualitative results of the Pix2Pix model are shown in Fig. 6.

XI. Evaluation Metrics and Comparative Analysis

The performance of the proposed models is quantitatively evaluated using two widely adopted image quality assessment metrics, namely Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM). These metrics assess the similarity between the generated image and the corresponding ground truth image.

PSNR measures the reconstruction quality of the generated image by comparing the pixel-wise differences between the generated and reference images. Higher PSNR values indicate better image reconstruction quality.

$$PSNR = 10 \log_{10} \left(\frac{MAX^2}{MSE} \right) \tag{2}$$

where MAX represents the maximum possible pixel value of the image and MSE denotes the Mean Squared Error between the generated and reference images.

SSIM evaluates the perceptual similarity between two images by considering luminance, contrast, and structural information. The value of SSIM ranges from 0 to 1, where a value closer to 1 indicates higher structural similarity.

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \tag{3}$$

where μ_x and μ_y denote the mean intensities of the two images, σ_x^2 and σ_y^2 represent their variances, and σ_{xy} is the covariance between the images.

TABLE III: Average Performance of the Proposed Pix2Pix Model on the LOL Test Set

Evaluation Metric	Average Value
PSNR (dB)	17.3511
SSIM	00.6396

XII. Conclusion

In this work, a deep learning-based approach for generating daylight images from low-light images using the Pix2Pix Generative Adversarial Network (GAN) was proposed. The experimental results demonstrate that the model effectively improves image brightness, visibility, and color consistency while preserving important structural details.

The performance of the proposed model was evaluated using qualitative analysis and quantitative metrics such as PSNR and SSIM. The generated results indicate that Pix2Pix can successfully translate low-light images into visually realistic daylight images.

Overall, the proposed approach provides an effective solution for low-light image enhancement and can be extended to applications such as surveillance systems, autonomous driving, and image restoration.

XIII. Future Work

Although the proposed Pix2Pix model produces promising results for low-light image enhancement and daylight image generation, there are several directions for future improvement. Larger and more diverse datasets can be used to improve the generalization capability of the model under different lighting conditions and real-world scenarios. Advanced architectures incorporating attention mechanisms and transformer-based networks can also be explored to enhance image quality and preserve finer details.

Future work may focus on reducing artifacts in extremely dark regions and optimizing the model for real-time deployment on mobile and embedded devices. The proposed approach can also be extended to applications such as video enhancement, surveillance systems, and autonomous driving.

XIV. References

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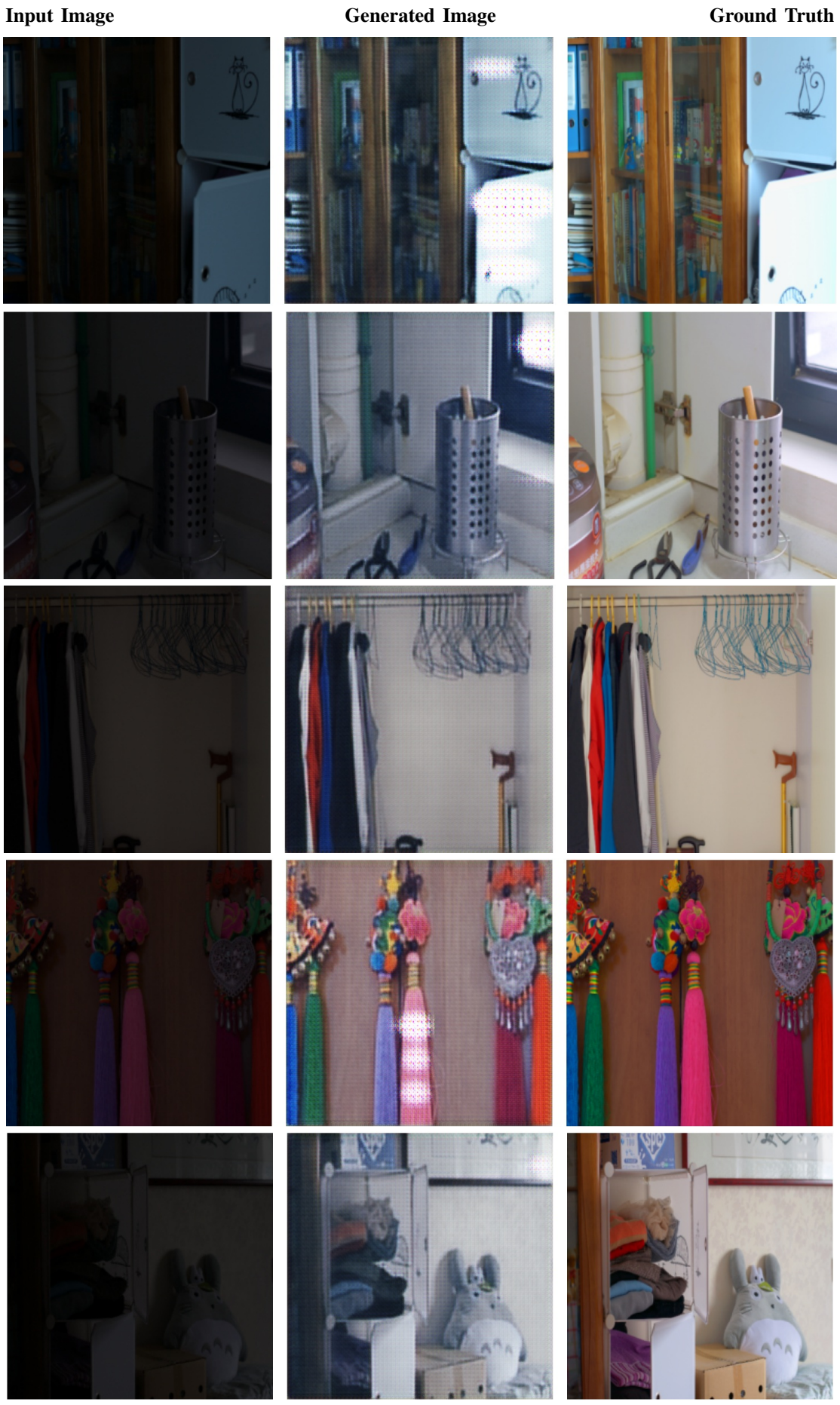
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Input Image
Generated Image
Ground Truth

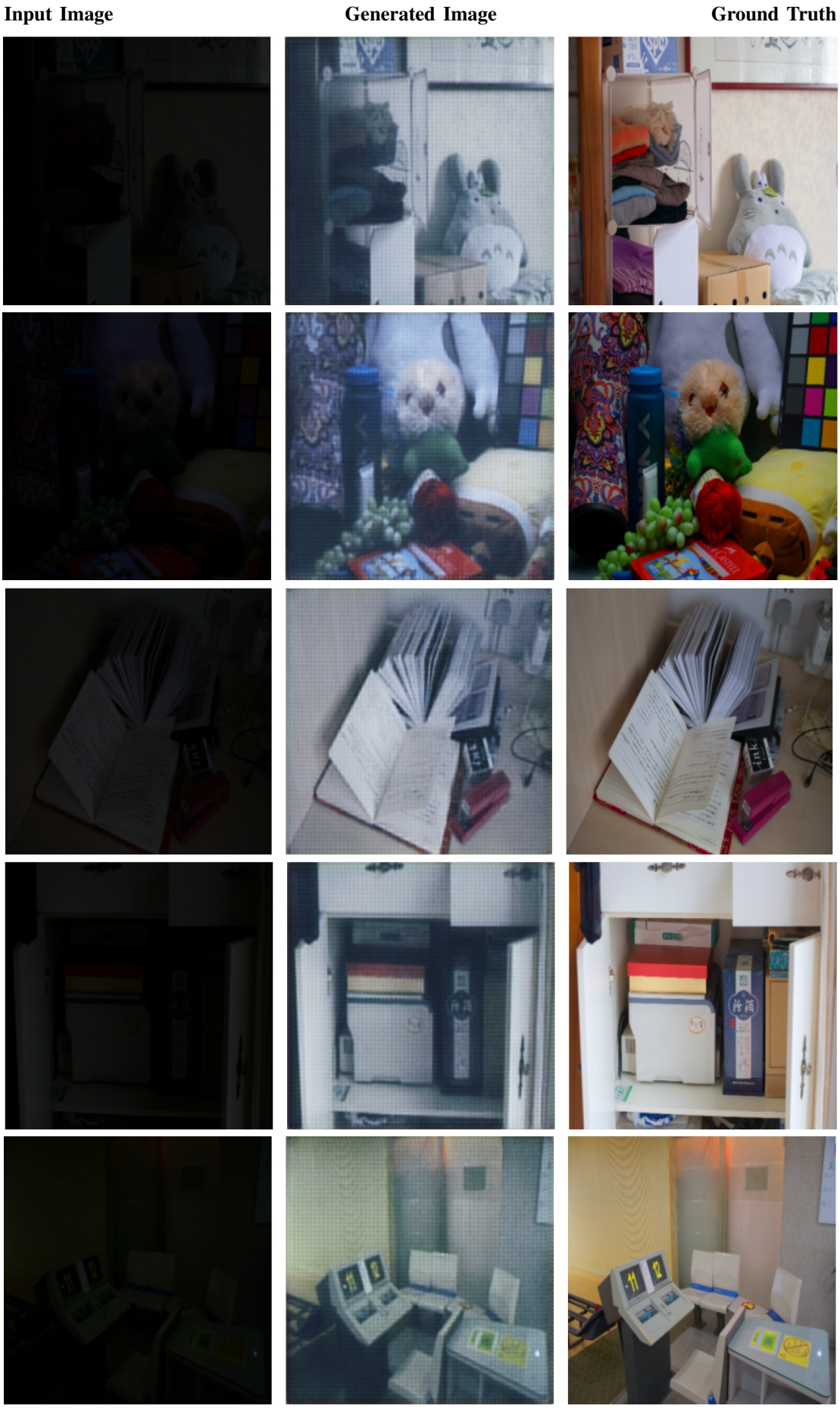


Fig. 6: Qualitative results of the Pix2Pix model.